

EC600M Series

Reference Design

LTE Standard Module Series

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Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

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About the Document

Revision History

Version	Date	Author	Description
-	2022-03-17	Hubery HUANG	Creation of the document
1.0	2022-07-12	Hubery HUANG	First official release
1.1	2023-02-24	Howell KANG	1. Changed the voltage domain of the ADC from 1.8 to 1.2 V (Sheet 1 & Sheet 3). 2. Updated the UART Level-shifting Circuit in IC Solution (Sheet 8).
1.2	2024-10-25	Kelly WANG/ Ronter ZHANG	1. Added applicable modules EC600M-EU and EC600M-LA. 2. Updated the power supply for module to 1.5 A (Sheets 1 & 2 & 5). 3. Added pins 130–135 of EC600M-EU and EC600M-LA (Sheet 3). 4. Specified that USIM2 is optional function (Sheets 1 & 3 & 7). 5. Updated antenna interface design and added relevant notes (Sheet 6). 6. Added a note about UART hardware flow control design (Sheet 8).

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1 Reference Design

1.1. Introduction

This document provides the reference design for Quectel EC600M series (comprising EC600M-CN, EC600M-EU and EC600M-LA) modules, including block diagram, power system block diagram, module interfaces, power supply design, antenna interface, USIM interfaces and UART interface.

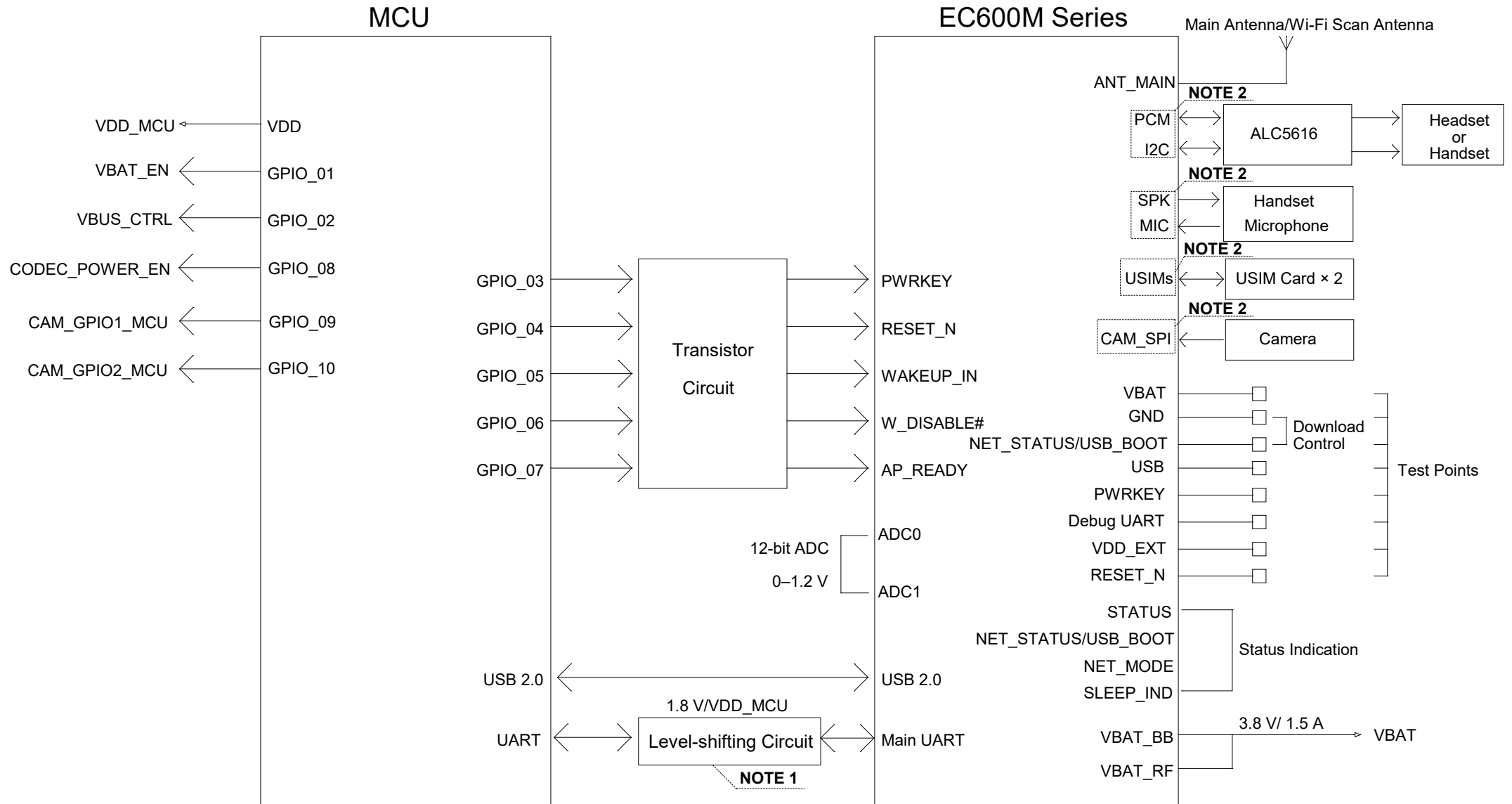
1.2. Schematics

The schematics illustrated in the following pages are provided for your reference only.

NOTE

It is required to confirm the applicability and price from the supplier about the IC involved in the reference design.

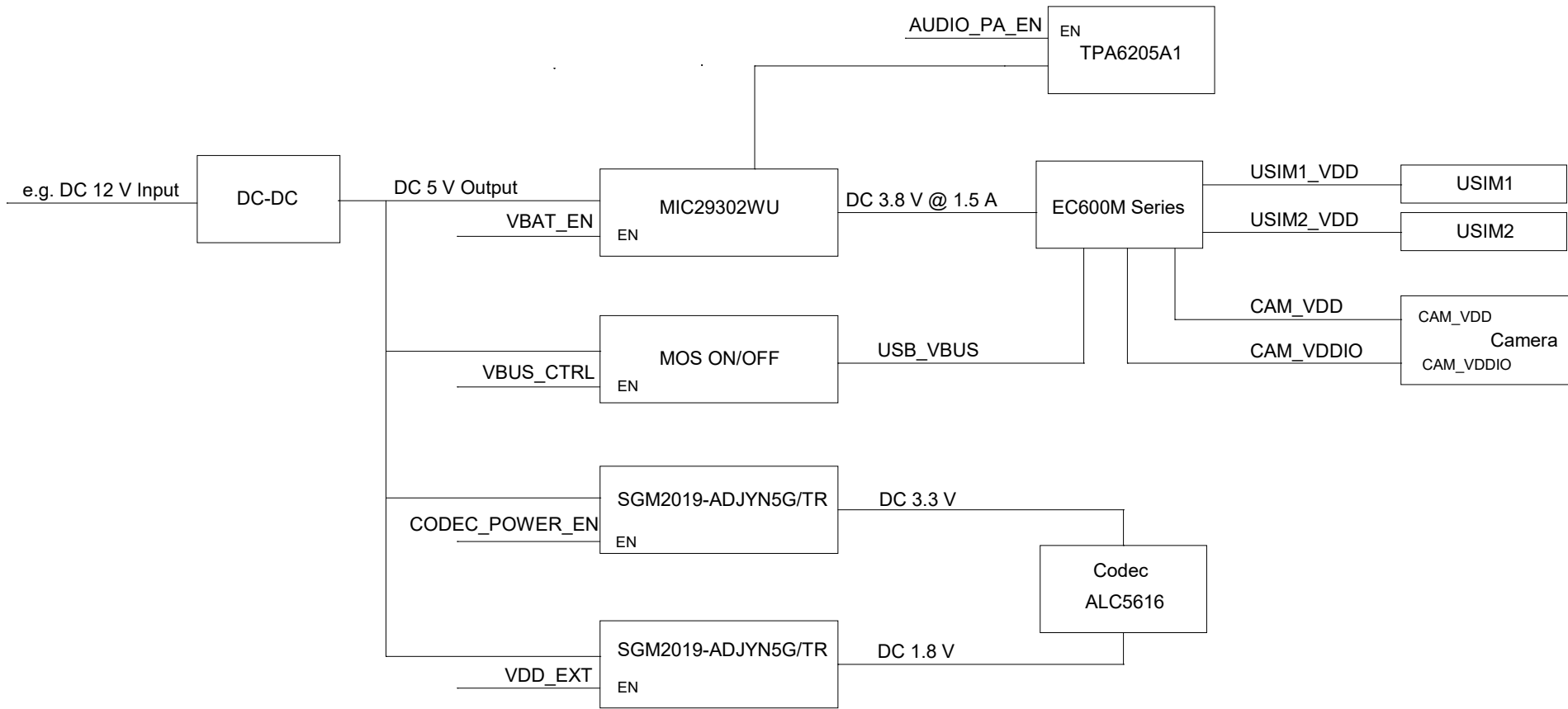
Block Diagram



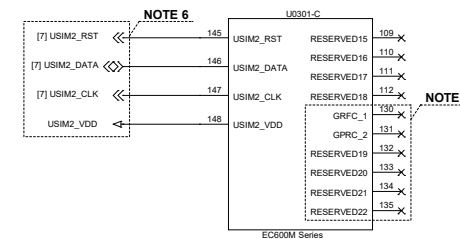
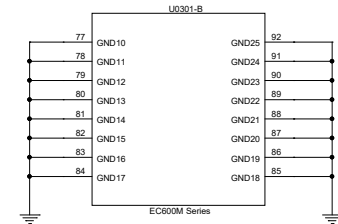
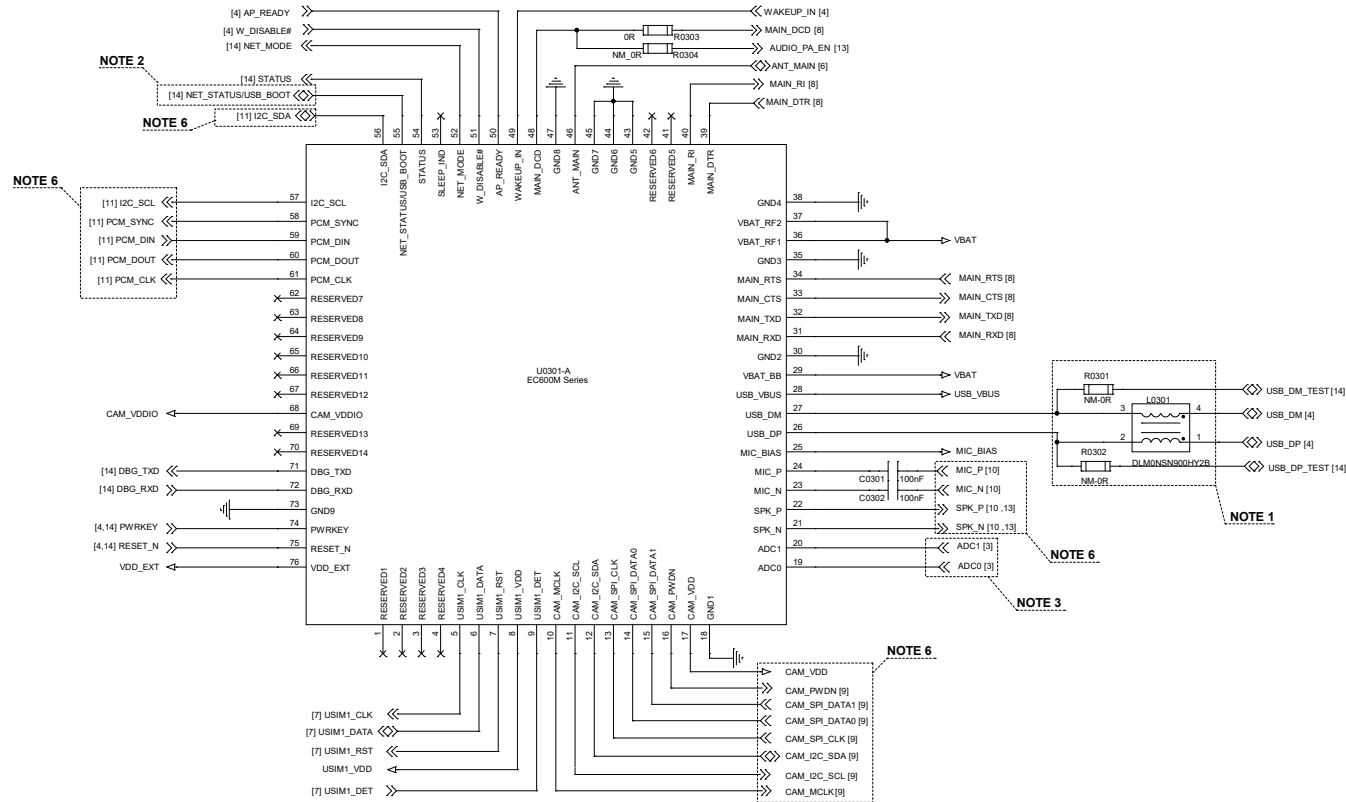
NOTE:

1. A transistor solution or an IC solution TXS0108EPWR provided by Texas Instruments is recommended.
2. PCM, I2C, USIM2, camera and analog audio functions are optional.

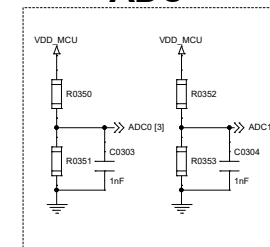
Power System Block Diagram



Module Interfaces



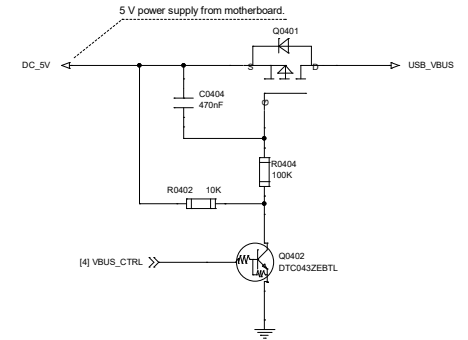
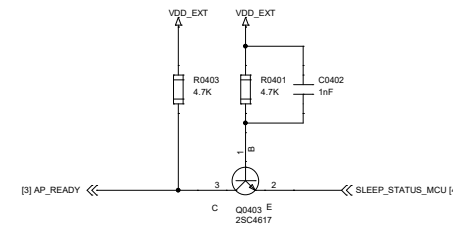
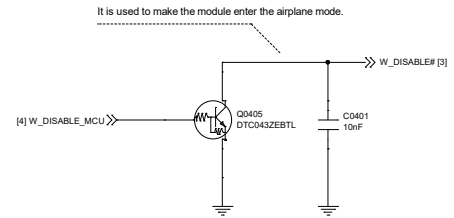
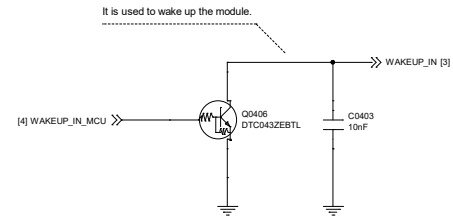
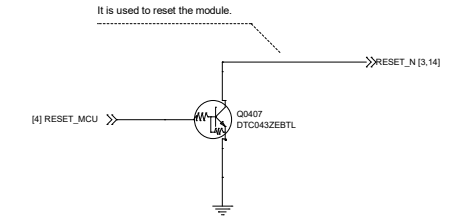
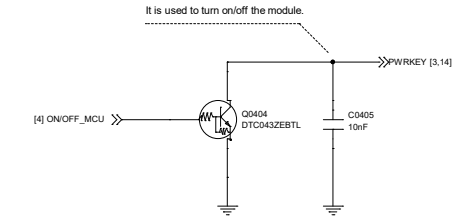
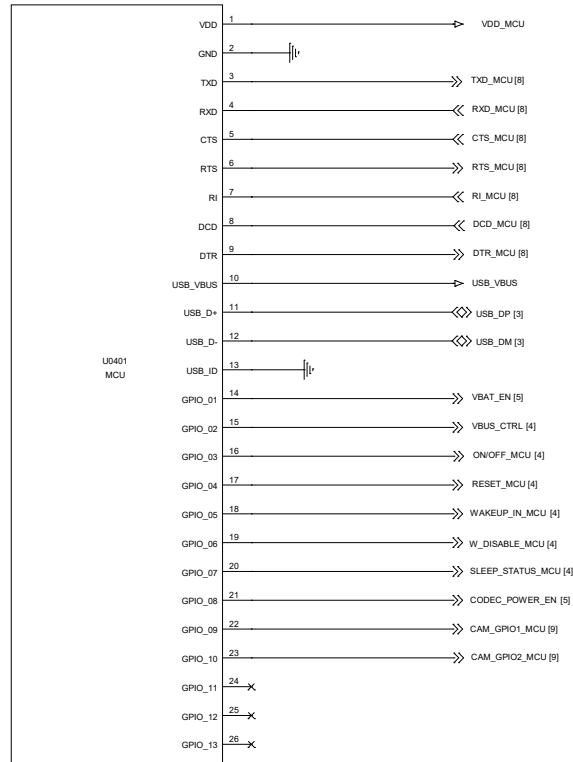
ADC



NOTE:

- It is recommended to add a common mode choke L0301 in series between the module and your MCU to suppress EMI. Additionally, test points must be reserved over USB_DP and USB_DM for firmware upgrades, and it is recommended to minimize extra trace stubs. Place L0301 and two resistors, R0301 and R0302, close to the module to ensure USB signal integrity.
- NET_STATUS/USB_BOOT cannot be pulled down to low level before the module starts up successfully.
- The input voltage range of ADC0 and ADC1 is 0–1.2 V. A voltage divider circuit with two resistors must be used for ADC0 and ADC1 voltage inputs respectively, and the required resistance of the two resistors (R0350 and R0352) that connected to VDD_MCU is between 100 kΩ and 1MΩ. The accuracy of the resistors in each voltage divider directly affects ADC sampling error. It is recommended to use resistors with an accuracy of 1 %. For higher ADC accuracy, resistors with accuracy of 0.5 % are recommended.
- Connect all GND pins to ground, and keep unused and RESERVED pins unconnected.
- Ensure an uninterrupted reference ground plane below the module, with minimal distance between the ground plane and the module layer. At least four-layer board design is recommended.
- PCM, I2C, USIM2, camera and analog audio functions are optional.
- The 6.0 and above version of QFlash tool must be used for firmware upgrading.
- EC600M-CN does not have pins 130–135.

MCU Interface



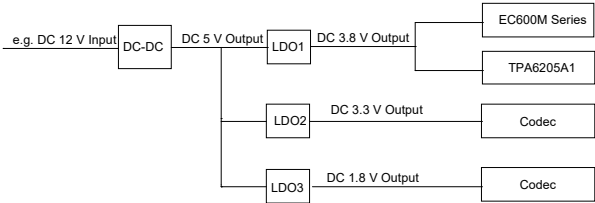
NOTE:

1. If the power domain of your MCU (U0401) is also 1.8 V, the level-shifting circuit is not necessary as it matches the 1.8 V power domain of the module's GPIO interfaces.
2. The USB interface of the module only serves as a slave device and supports full-speed and high-speed modes of USB 2.0. To communicate with the USB interface, MCU needs to support USB host mode or OTG function.
For USB detection, the USB_VBUS pin of the module should be powered by an external power system. Use VBUS_CTRL to control the on/off state of the USB_VBUS power supply.
3. It is recommended to choose MCU GPIO pins with a default low level to control the module's PWRKEY and RESET_N pins. Ensure that the load capacitance on these pins does not exceed 10 nF.

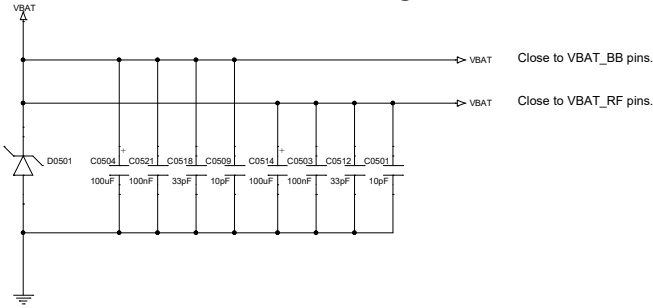
Power Supply Design

DC-DC Application

When the input voltage is above 7.0 V, use a DC-DC converter to convert the high input voltage to 5.0 V, and then use LDOs to convert it to 3.8 V, 3.3 V and 1.8 V to power the module, audio PA and Codec.

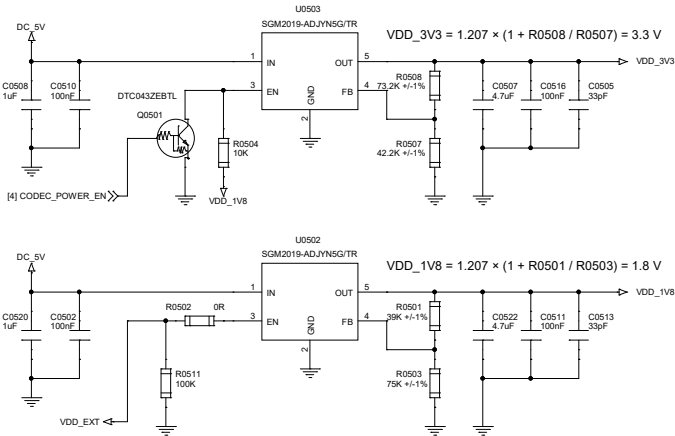


VBAT Design



- NOTE:**
1. The power supply for the module should be capable of supplying a minimum current of 1.5 A.
 2. The VBAT trace should be connected to VBAT_BB and VBAT_RF pins in a star configuration.
 3. The width of VBAT_BB trace should be at least 1 mm; and the width of VBAT_RF trace should be at least 2 mm.
 4. The recommended operating voltage range for VBAT is 3.4 V to 4.3 V, with a typical value of 3.8 V.

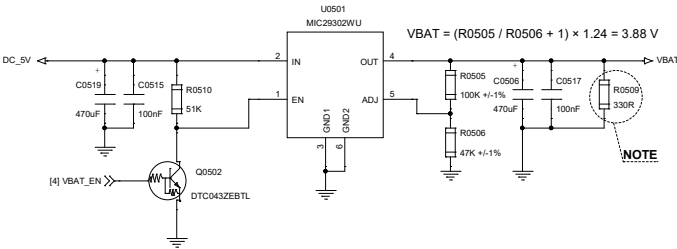
Power Supply for Codec



- NOTE:**
1. VDD_EXT and CODEC_POWER_EN are used to turn on/off VDD_1V8 and VDD_3V3 respectively.
 2. To ensure proper functioning of the audio codec, adhere to the following power-up/down sequences:
Power-up sequence: power on VDD_1V8 first, followed by VDD_3V3.
Power-down sequence: power off VDD_3V3 first, followed by VDD_1V8.

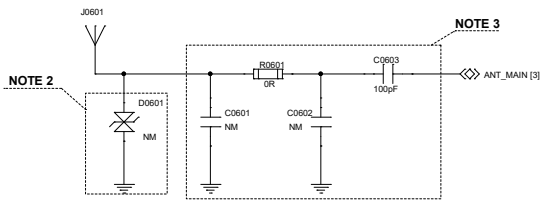
LDO Application

When the input voltage is below 7.0 V, use an LDO to convert the input voltage to 3.8 V.



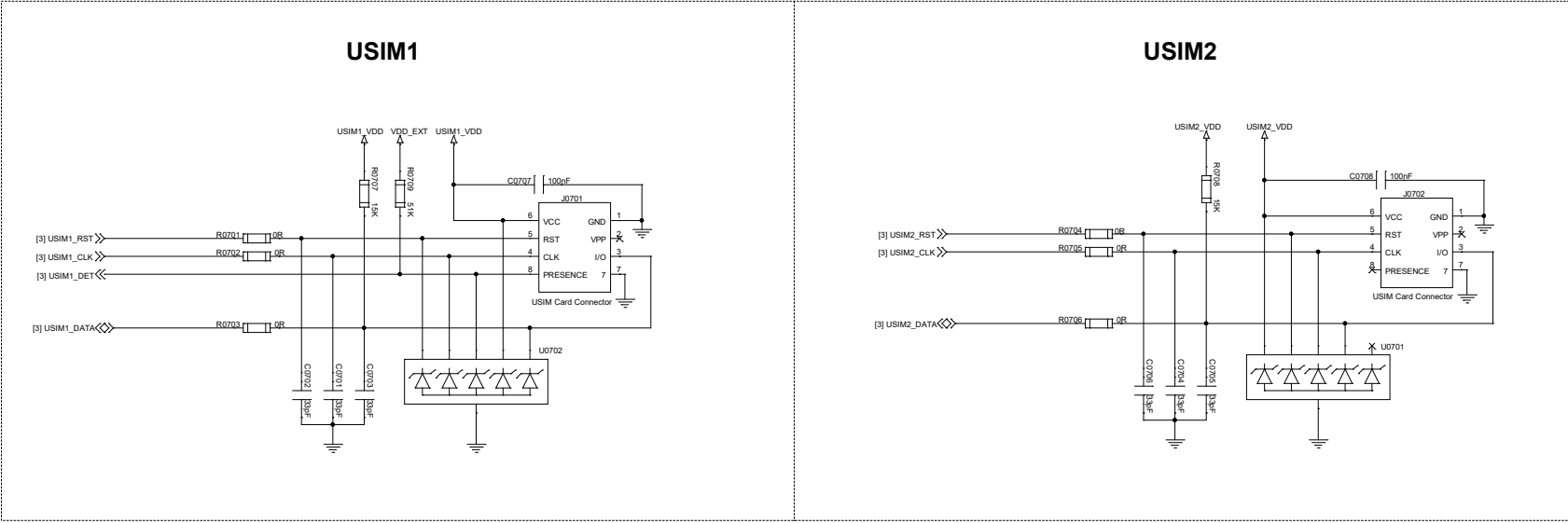
- NOTE:**
- The recommended load current should exceed 10 mA.

Antenna Interface Design



- NOTE:**
- 1. The single-ended impedance of the RF antenna is 50 Ω.
 - 2. It is recommended to reserve an ESD protection component for the antenna interface and the junction capacitance should not exceed 0.05 pF.
 - 3. Reserve a dual L-type circuit at antenna interface. If there is DC power at the antenna ports, C0603 must be used for DC-blocking to prevent short circuit to ground.
The capacitance value is recommended to be 100 pF, which can be adjusted according to actual requirements. If there is no DC power in the peripheral design, C0603 should not be reserved.
 - 4. The module supports Wi-Fi Scan function. Wi-Fi Scan function utilizes the same antenna interface as the main antenna. Due to this shared interface, Time Division Multiplexing (TDM) is employed since the two functions cannot be used simultaneously.

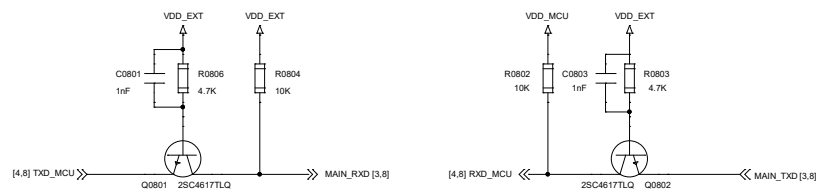
USIM Interface Design



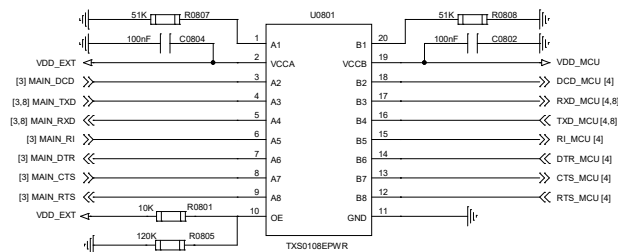
- NOTE:**
1. It is recommended to use U0701 and U0702 for effective ESD protection with a parasitic capacitance below 15 pF.
 2. For USIM_DATA, it is recommended to add a 15 kΩ pull-up resistor R0707 and R0708 near the (U)SIM card connector to improve the anti-jamming capability of the (U)SIM card.
 3. R0701–R0706 are used for debugging, and C0701–C0706 used for filtering out RF interference should be placed close to the USIM card connector.
 4. The capacitance of C0707 and C0708 should not exceed 1 μF and they should be placed close to the USIM card connector.
 5. USIM2 is optional. If the function is required, contact Quectel Technical Support.
 6. Only USIM1 supports hot-plug function.

UART Interface Design

UART Level-shifting Circuit - Transistor Solution

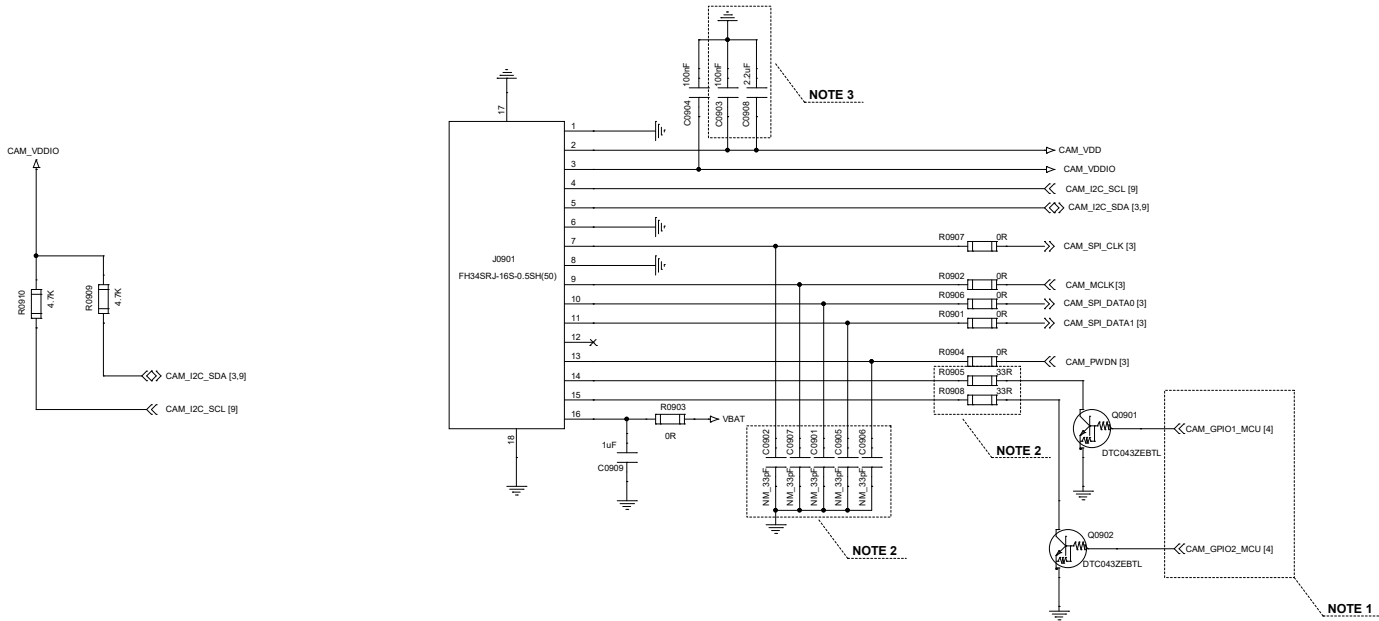


UART Level-shifting Circuit - IC Solution



- NOTE:**
- There are two level-shifting solutions: transistor solution and IC solution, and it is recommended to select the latter one.
 - The power supply of TXS0108EPWR's VCCA should not exceed that of VCCB. For more information, please refer to the datasheet of TXS0108EPWR.
 - The transistor solution is not suitable for applications with high baud rates exceeding 460 kbps. The capacitors C0801 and C0803 of 1 nF can improve the signal quality.
 - MAIN_CTS and MAIN_DTR level-shifting circuits are similar to that of the MAIN_RXD interface.
MAIN_CTS, MAIN_RI and MAIN_DCD level-shifting circuits are similar to that of the MAIN_TXD interface.
 - To increase the stability of UART communication, it is recommended to add UART hardware flow control design.

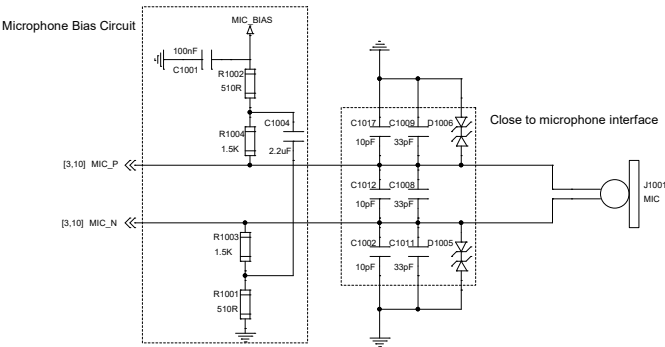
Camera Interface Design



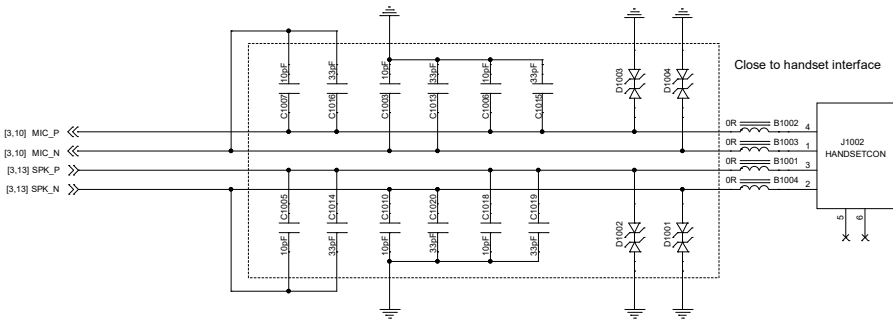
- NOTE:**
1. To control the cathode of the camera's positioning light, use CAM_GPIO1_MCU with a triode switching circuit. Similarly, CAM_GPIO2_MCU controls the cathode of the camera's supplement light. It is recommended to choose GPIO pins that are in a pull-down state by default as these two control pins.
 2. Reserve 33 pF capacitors for the signal pins and use them for debugging. The values of current limiting resistors (R0905 and R0908) for the positioning light and supplement light should be adjusted based on the desired brightness level.
 3. Connect the capacitors (C0903 and C0908) of the CAM_VDD power supply directly to the GND layer. Failure to do so may result in power supply noise causing abnormalities such as white dots on the preview screen.
 4. The camera function is optional.

Analog Audio Design

Microphone Application



Handset Application

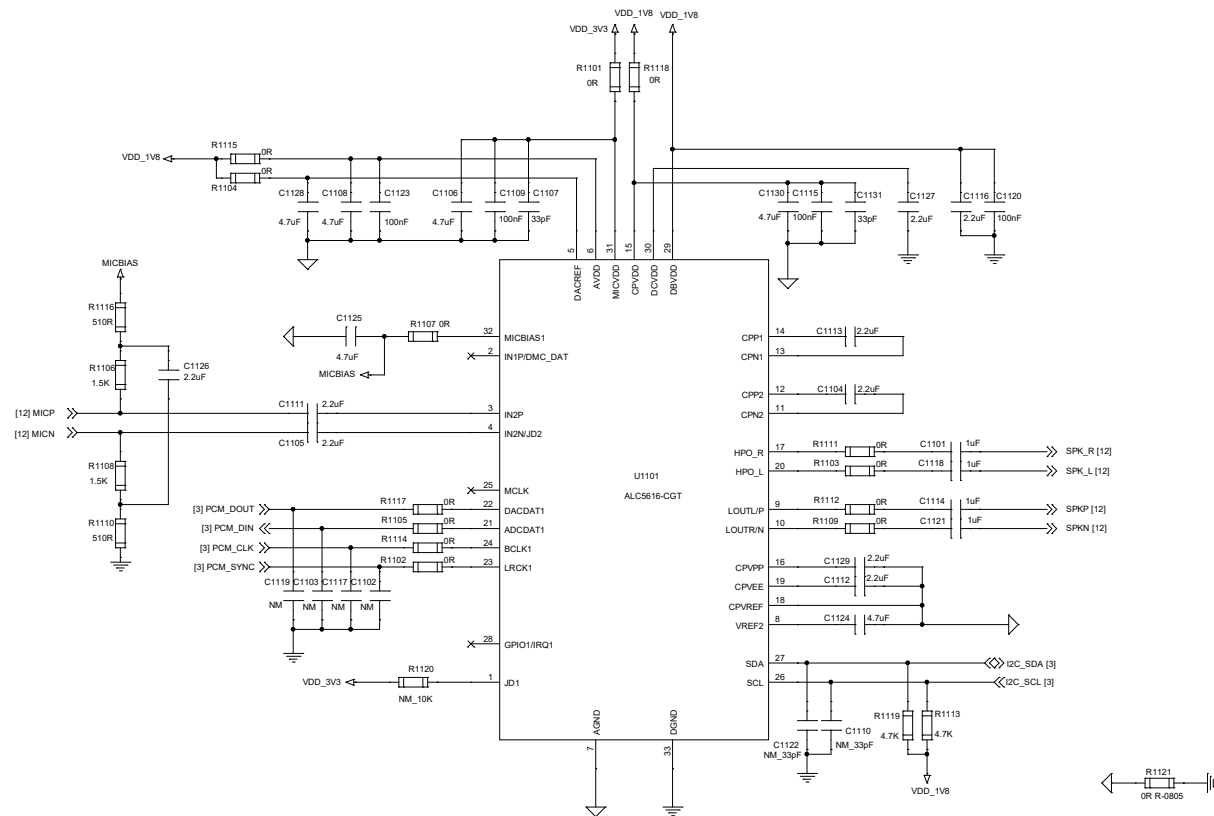


NOTE:

- Both the MIC and SPK signal traces need to be routed as differential pairs.
- Surround all MIC and SPK signal traces with ground on the same layer and with ground planes above and below to minimize noise interference.
- In the audio design, you can choose either the analog audio or the codec. It is not necessary to select both.
- It is recommended to use 10 pF and 33 pF capacitors to filter RF interference.
- The analog audio function is optional.

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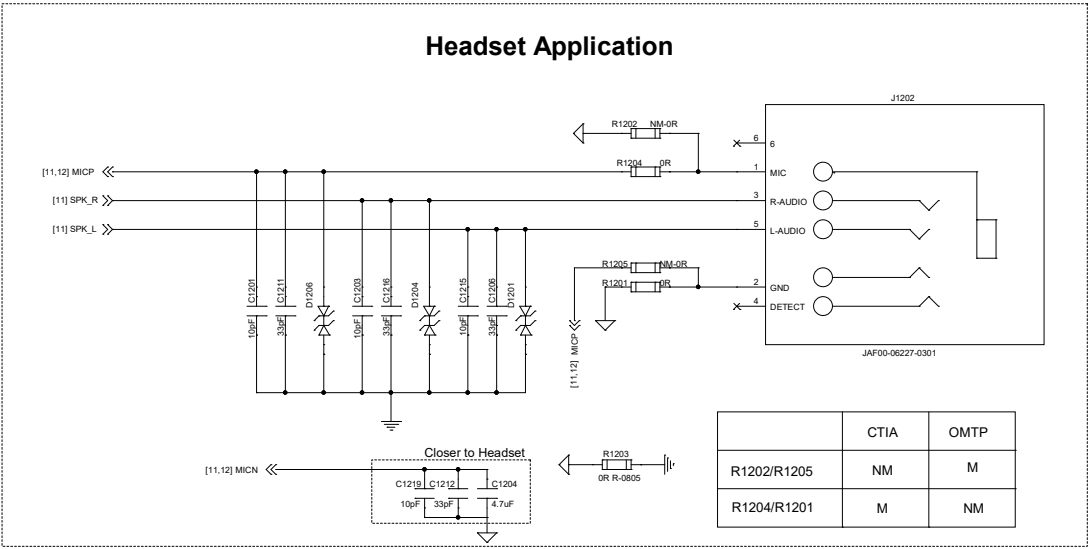
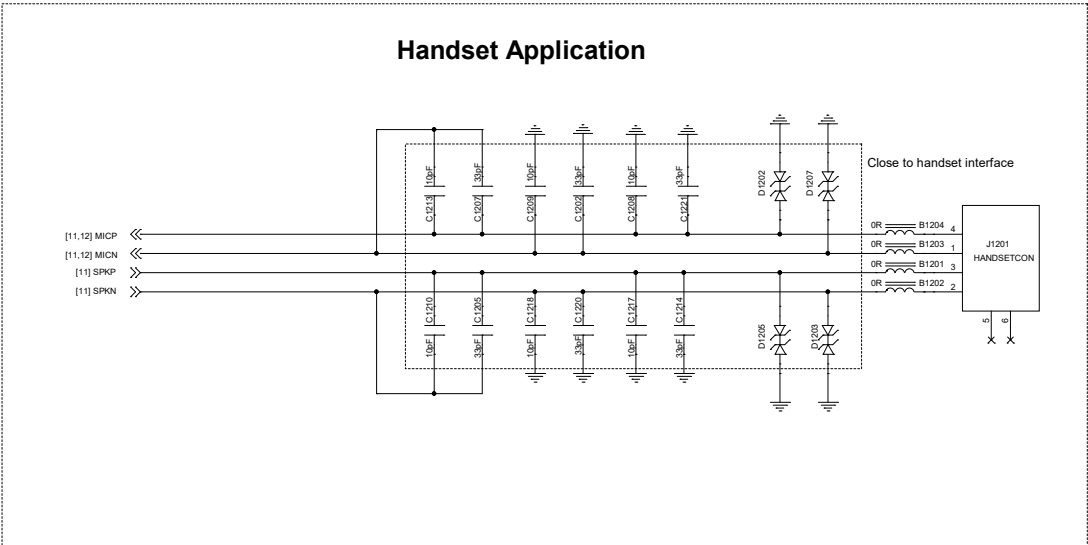
Audio Codec Design (ALC5616)



NOTE:

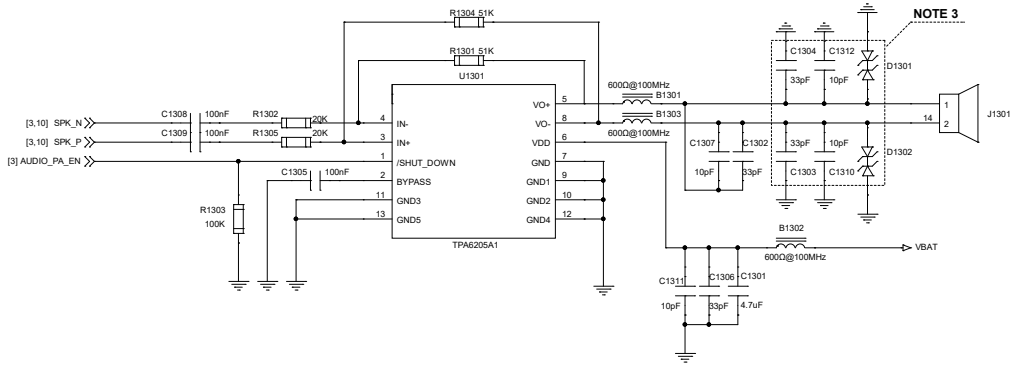
1. ALC5616 power-up sequence: DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD → MICVDD → software initialization.
2. ALC5616 power-down sequence: disable Codec function by software → MICVDD → DBVDD/I2C pull-up power/AVDD/DACREF/CPVDD.
3. Ensure all power supplies for the codec are powered on before the module automatically initializes the Codec via the I2C interface.
4. Differentiate between analog ground and digital ground. Connect analog ground and digital ground using a 0 Ω resistor (R-0805). Refer to "Audio Codec Interface Design" for more details.
5. For more details, please refer to the datasheet of ALC5616.
6. The analog audio function is optional.

Audio Codec Interface Design



- NOTE:**
1. The Codec analog output can drive handset and headset. For larger power loads such as loudspeaker, add an audio power amplifier in the design.
 2. In handset application, route the MIC and SPK signal traces as differential pairs.
 3. In headset application, route the MIC signal traces as a differential pair.
 4. Surround all MIC and SPK signal traces with ground on the same layer and with ground planes above and below to minimize noise interference, such as clock and DC-DC signals.
 5. Differentiate between analog ground and digital ground. Analog ground should have a direct via to digital GND through a 0 Ω resistor (R-0805).
 6. The analog audio function is optional.

Analog Audio Design (Audio Power Amplifier)

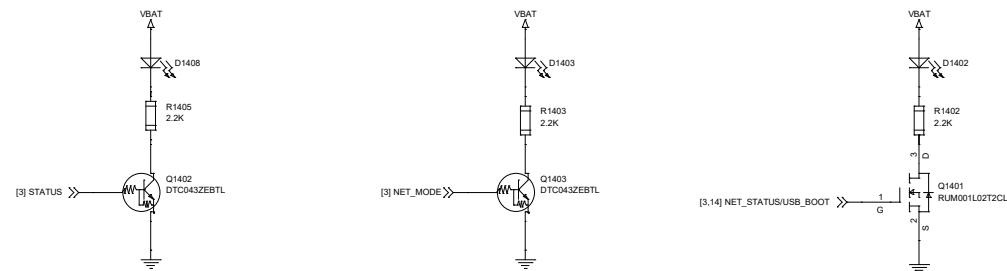


NOTE:

1. SPK_P and SPK_N channels are differential output channels intended for connecting to an external audio power amplifier.
To eliminate Pop noise, it is recommended to utilize MAIN_DCD of the module as the control signal for the audio power amplifier's enable pin.
For more information about AUDIO_PA_EN, please contact Quectel Technical Support.
2. The type of power amplifier in this design is for reference only. Select the appropriate audio power amplifier according to actual needs.
3. When designing the layout, ensure that filter capacitors and ESD protection components are placed close to the loudspeaker to filter out interference and provide adequate protection.
4. The selection of ESD protection components should consider the output voltage range of the audio power amplifier.
Ensure that the output voltage of the amplifier remains within the maximum reverse working voltage range of the selected ESD protection components under normal operating conditions. This precaution helps prevent damage to the ESD protection components.
5. The analog audio function is optional.

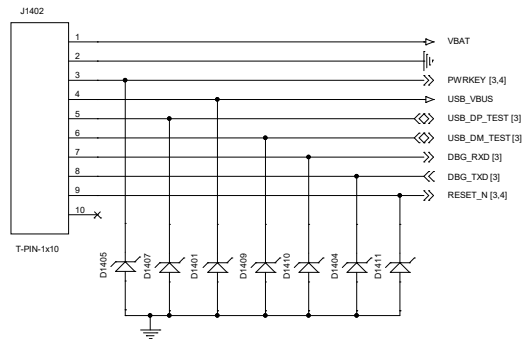
Other Designs

Indicators



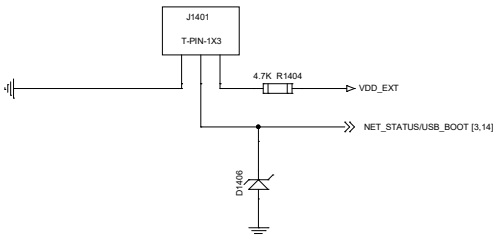
- NOTE:**
- For more details about STATUS, NET_MODE and NET_STATUS/USB_BOOT, please refer to the hardware design document of the module.
 - To minimize the module's power consumption during the sleep mode of your device, replace the power supply (VBAT) of the STATUS, NET_MODE and NET_STATUS/USB_BOOT indicators with externally controllable sources and turn off the indicators when the module is in sleep mode.
 - Note that the maximum value of the MOSFET Q1401's $V_{gs(th)}$ should not exceed 1 V, since the NET_STATUS/USB_BOOT pin of the module outputs high level by default.
If a transistor such as Q1402 and Q1403 is used to replace the MOSFET, the NET_STATUS/USB_BOOT pin will be pulled down to low-level, and the module will enter forced download mode and cannot be turned on normally.

Reserved Test Points



- NOTE:**
- Test points for both USB and debug UART interfaces are reserved for capturing logs.
 - Test points for USB interface can also be reserved for firmware upgrading.
 - The junction capacitance of the ESD protection components on USB data traces should be less than 2 pF.
 - The debug UART interface supports a 1.8 V power domain. If your application operates at 3.3 V, use a voltage-level translator.

USB_BOOT Interface



- NOTE:**
- Make sure to reserve the USB_BOOT interface design and it is recommended to reserve a test point for NET_STATUS/USB_BOOT.
 - Before turning on the module, pull NET_STATUS/USB_BOOT down to GND to activate the forced download mode.
This mode enables firmware upgrades via the USB interface.
 - The 6.0 and above version of QFlash tool must be used for firmware upgrading.

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